

Information Overload, Cognitive Capacity, and Customer Engagement: Evidence from Retail Messaging Data

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Abstract

We study how attention constraints and messaging fatigue shape customers' decisions to open promotional messages, and what this implies for firms' communication strategies. Using high-frequency engagement data and a machine-learning framework, we analyze heterogeneity in responses to message frequency and content across dormant, occasional, and loyal customers. Customer segments are constructed using proxies for prior beliefs and prior precision, capturing persistent differences in information processing.

Our results reveal strong non-linear and heterogeneous fatigue effects. While moderate exposure can increase engagement, higher message volume is associated with increasingly negative engagement responses, particularly for customers with lower prior precision. Model-agnostic interpretability analysis indicates that cognitive load affects engagement through both the evaluation of message signals and the effective cost of opening. Counterfactual simulations illustrate trade-offs between messaging intensity and effectiveness and suggest that reductions in aggregate message pressure can disproportionately benefit less engaged customers. Although the analysis relies on proxies for latent cognitive mechanisms and focuses on openings as the primary outcome, the findings highlight the central role of attention constraints in shaping sustainable communication strategies in digital markets.

1 Introduction

In modern digital markets, customers are repeatedly contacted by firms across multiple communication channels. A single notification feed can contain dozens of promotional messages from different senders within a few days, and even messages from genuinely interesting sources risk being lost in the noise. Opening a message is not automatic: the customer must notice the notification, form an expectation that the content is worth reading, and spend time and cognitive effort to process it. As the volume of incoming messages grows, customers can feel overloaded, and ignoring or deleting messages becomes the default response.

These observations motivate our central economic questions:

1. How do attention constraints and features of the information environment shape the probability that a customer opens a promotional message?
2. How should firms adjust messaging strategies in response to these constraints?

In our framework, consumers treat each incoming message as a noisy signal about its potential value and decide whether to open it by comparing expected informational gains to the cognitive cost of processing. At the moment of arrival, customers observe only limited cues—such as notification text, subject lines, visible keywords, timing, or sender identity—rather than the full content of the message. When prior beliefs about the sender are pessimistic or cognitive load is high, the perceived benefit of paying attention may fall below this cost, making inattention an optimal response rather than a mechanical mistake. Fatigue plays a dual role: it raises the cost of attention and weakens how effectively customers evaluate observable cues, reducing their effective informational value.

We interpret these mechanisms using insights from rational inattention and Bayesian persuasion. Rational inattention models emphasize that individuals face limited information-processing capacity and must allocate attention selectively (Sims, 2003; Matějka & McKay, 2015). In our setting, opening a message is a small but costly attention decision, and recent exposure or notification congestion effectively raises its marginal cost. Persuasion-oriented models highlight how senders can influence receiver behavior through signal design (Kamenica & Gentzkow, 2011). Rather than modeling fully dynamic optimization or optimal message sequences, we focus on how repeated exposure and simple, observable message features shape engagement under attention constraints.

Our approach is also closely related to sparsity-based models of bounded rationality, in which decision-makers deliberately ignore many information dimensions (Gabaix, 2014). We interpret message-opening behavior as a sparse attention rule: customers attend only to a limited subset of senders or message features perceived as salient, while treating the remainder as noise. As fatigue accumulates, this attentive set shrinks further, increasing the likelihood of persistent disengagement even when messages retain potential value.

Contributions. This paper makes three contributions. First, it provides an empirically grounded analysis of how attention constraints and messaging fatigue shape engagement decisions in high-frequency communication environments. Second, using large-scale campaign data and flexible machine-learning methods, it documents substantial heterogeneity in responses to message frequency, content cues, and congestion across loyal, occasional, and dormant customers. Third, it links reduced-form predictive evidence to a microeconomic framework of attention and persuasion, using model-based interpretation tools and counterfactual comparisons to derive implications for sustainable messaging strategies under limited attention.

2 Related Literature

Our paper draws on three main strands of literature. First, rational inattention models treat attention as a scarce resource that agents allocate optimally. Sims (2003) formalizes information-processing constraints, while Matějka & McKay (2015) show how such constraints can generate discrete choice behavior resembling standard logit models. We apply these ideas to messaging environments by interpreting message openings as attention choices whose cost increases with recent exposure and fatigue.

Second, we relate our analysis to the Bayesian persuasion literature, which studies how a sender can influence a receiver’s actions through information design (Kamenica & Gentzkow, 2011). While much of this literature focuses on static or stylized dynamic settings, recent work considers environments with costly listening and disengagement (Che and Mierendorff, 2019; Che et al., 2022).

We draw on these insights to motivate how repeated messaging and accumulated exposure affect attention incentives, without modeling optimal dynamic message sequences.

Finally, our work is connected to bounded rationality and sparse attention models (Gabaix, 2014). These models emphasize how decision-makers simplify complex information environments by focusing on a limited number of salient cues. We contribute by documenting how such sparse attention patterns appear empirically in large-scale messaging data and by showing how heterogeneity in attention sensitivity shapes engagement across customer segments.

3 Economic Framework

We study customer decisions to open promotional messages, including emails and push notifications. Each message is treated as a noisy signal about its relevance. A customer opens a message if the expected benefit from processing it exceeds the cognitive cost of doing so. Formally,

$$a_{it} = \begin{cases} 1 & \text{if } EV_{it} - C_{it} \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

where EV_{it} is the expected value of opening and C_{it} is the cost of attending to and processing the message at time t .

The framework combines Bayesian updating with rational inattention and a state-dependent cognitive load process. It draws on Bayesian persuasion models (Kamenica & Gentzkow, 2011) and rational inattention approaches to attention allocation (Sims, 2003; Matějka & McKay, 2015), adapted to a setting with persistent cognitive fatigue as in (Che and Mierendorff, 2019; Che et al., 2022).

3.1 Agents, Timing, and Cognitive Load

Two agents interact: a focal firm and a customer. The firm chooses message frequency and observable design features such as timing, clarity, and personalization. The customer receives messages over time and decides whether to open each one, subject to limited attention and evolving cognitive load.

At time t , the sequence is as follows. Before the message arrives, the customer holds a prior belief about the relevance of messages from the firm. Upon arrival, the customer observes a preview (e.g., sender, subject line, timing) at zero opening cost. Given current cognitive load L_{it} , the customer determines how much attention to devote to processing this preview, which governs the informativeness of the signal used to update beliefs. Finally, the customer compares the expected value of opening to the cognitive cost and chooses whether to open.

Cognitive load summarizes fatigue, competing demands on attention, and inbox congestion. Higher load reduces the customer’s ability and willingness to process new information and raises the perceived cost of opening messages.

3.2 Decision Problem

The opening decision at time t is static, conditional on the current state:

$$U_{it}(1) = EV_{it} - C_{it}, \quad (1)$$

$$U_{it}(0) = 0, \quad (2)$$

with $a_{it} = 1$ if $U_{it}(1) \geq U_{it}(0)$. The remainder of the framework characterizes how beliefs, attention, and load jointly determine EV_{it} and C_{it} .

3.3 Beliefs, Signals, and Load-Attenuated Previews

Let θ_i denote the latent relevance of messages from the firm for customer i . Prior to period t , beliefs follow

$$\theta_i \sim \mathcal{N}(\mu_i, \tau_\theta^{-1}),$$

where μ_i is the prior mean and τ_θ the prior precision, capturing how stable and predictable past experiences with the firm have been.

Upon message arrival, the customer observes a raw preview signal

$$s_{it} = \theta_i + \eta_{it}, \quad \eta_{it} \sim \mathcal{N}(0, \tau_s^{-1}),$$

where τ_s reflects the baseline informativeness of visible message attributes. Cognitive load attenuates the perceived strength of this preview. The effective signal entering belief updating is

$$s_{it}^{\text{eff}} = w(L_{it})s_{it}, \quad w(L_{it}) = \exp(-\delta L_{it}), \quad \delta > 0.$$

High load thus reduces how strongly the preview is perceived, even before accounting for attention allocation.

3.4 Rational Inattention and Chosen Precision

Processing the preview is costly. Following Sims (2003), information extraction is constrained by limited cognitive capacity. For Gaussian signals, the mutual information between θ_i and the processed signal summarizes informativeness. Let κ_{it} denote the amount of information the customer chooses to extract, incurring marginal cost c_{it} .

The customer chooses κ_{it} to trade off reduced posterior uncertainty against information costs, as in Matějka & McKay (2015) and Che and Mierendorff (2019). The optimal information usage implies an effective signal precision

$$\tau_s^* = \begin{cases} \frac{1}{c_{it}} - \tau_\theta, & \text{if } \frac{1}{c_{it}} > \tau_\theta, \\ 0, & \text{otherwise.} \end{cases}$$

When information is inexpensive, the customer can extract a large share of the signal's available precision, allowing previews to meaningfully influence beliefs. When information is costly, only a limited fraction of the signal is processed, so belief updating relies more heavily on prior beliefs rather than on the current preview.

3.5 Expected Value and Cognitive Costs

Given limited precision, the posterior expected value of opening is

$$EV_{it} = \frac{\tau_\theta}{\tau_\theta + \tau_s^*} \mu_i + \frac{\tau_s^*}{\tau_\theta + \tau_s^*} s_{it}^{\text{eff}}.$$

Cognitive load affects both information processing and the action cost of opening. The marginal cost of information increases with load,

$$c_{it} = c_0 + c_1 L_{it} + c_2 L_{it}^2, \quad c_2 > 0,$$

while the opening cost is

$$C_{it} = \alpha_i + \beta_1 L_{it} + \beta_2 L_{it}^2.$$

Higher load therefore reduces responsiveness to message previews by degrading effective signals, raising information costs, and increasing the effort cost of opening.

3.6 Evolution of Load

Cognitive load evolves according to

$$L_{it} = \rho L_{i,t-1} + \gamma_1 MSG_{it} + \gamma_2 COMP_{it} + \gamma_3 EVENT_t + \gamma_4 TIME_t + \varepsilon_{it}^L,$$

where MSG_{it} captures the firm's messaging intensity, $COMP_{it}$ competing messages, and other terms reflect contextual factors. Persistence in load introduces state dependence without requiring customers to solve a fully dynamic optimization problem.

3.7 Key Assumptions in the Economic Framework

To clarify the scope of our theoretical model, we explicitly state the key assumptions underlying our framework:

1. **Bounded Rationality:** Customers optimize under limited attention and cognitive constraints, meaning they cannot process all available information perfectly.
2. **Rational Inattention:** Beliefs are updated based on signals, but the precision of these updates is constrained by attention costs.
3. **Signal Structure:** The value and precision of a marketing message depend on observable features such as subject line length, personalization, and promotional keywords.
4. **Fatigue Accumulation:** Cognitive load increases with message volume, generating non-linear, segment-specific responses in engagement.
5. **Segment Heterogeneity:** Customers differ in prior beliefs and attentional capacity, which is reflected in their historical engagement patterns.

These assumptions provide the foundation for the design of our proxies and the mapping to empirical features in subsequent analysis.

3.8 Testable Predictions

The economic framework generates several empirically testable patterns regarding how messaging intensity, content, and customer heterogeneity affect opening behavior:

1. **Fatigue and attention effects:** Higher recent messaging volume and load should reduce the probability of opening a message, with the effect strongest for customers with lower prior precision or weaker engagement history.
2. **Signal informativeness:** Messages with more informative previews (e.g., clear subject lines, personalized content) should have larger effects on opening probabilities, particularly under low-to-moderate load, while high-load states attenuate this impact.
3. **Non-linear exposure response:** Moderate increases in message frequency can increase engagement for some customers, but excessive frequency may generate diminishing or negative returns due to rising cognitive costs.
4. **Heterogeneity across customer segments:** Loyal, occasional, and dormant customers are expected to respond differently to message frequency and content, reflecting differences in priors, precision, and sensitivity to load.

These predictions provide a bridge to the empirical analysis, where high-frequency engagement data and machine-learning methods are used to characterize customer responses and heterogeneity.

4 Data & Methods

We utilize a publicly available dataset of anonymized marketing data from a large retail firm ¹. The data include messages sent via email and push notifications between April 2021 and March 2023. Each observation corresponds to a unique message delivered to a client, with information on whether the client opened it, the campaign type, subject line features, and temporal characteristics such as sending time.

Because the full dataset is very large (over 700 million messages), we draw a subsample of approximately 1.56 million messages sent to 35,000 users. This subsample preserves key metrics of the overall population, including open rates, timing patterns, and message characteristics.

The data is fully anonymized, so it lacks information about client characteristics and the product types offered to them.

4.1 Mapping Theory to Data

To link our empirical analysis to the economic framework, we construct proxies for the latent constructs described in Section 3. Each observed opening a_{it} is generated by the latent expected value of the message EV_{it} and the latent cognitive cost C_{it} , subject to idiosyncratic shocks and measurement errors:

$$a_{it} = \mathbf{1}(EV_{it} + \epsilon_{it}^{EV} \geq C_{it} + \epsilon_{it}^L),$$

¹See the public Kaggle dataset at <https://www.kaggle.com/datasets/mkechinov/direct-messaging/data>.

where ϵ_{it}^{EV} captures message- and context-specific deviations in expected value, and ϵ_{it}^L captures unobserved fluctuations in cognitive load. Measurement errors arise because our proxies are imperfect indicators of latent constructs.

We engineer features capturing:

1. **Prior belief and precision:** historical open rates and variance in opens over a rolling 14-day window, reflecting persistent beliefs and uncertainty.
2. **Signal value and precision:** subject line length and bonus keywords capturing message informativeness.
3. **Load and cognitive cost:** recent messaging volumes (7-day and 30-day windows), time-of-day/week indicators, simulated competitors, and global open rates capturing event-type fluctuations.

Across all engineered features, we face measurement challenges.² Our prior precision relies on the user’s variance in previous open rates, which itself may vary a lot if the sample size of the users’ previous messages is small. Other features such as load proxy capture only a subset of the true intended effect, and there may be a delay or overlap in high intensity periods (such as users receiving messages immediately after purchasing). Lastly, lack of specific information in regards to message specific and products creates challenges in identifying a clear boundary between a user who stops opening from fatigue, rather than simple disinterest. As a result, our empirical strategy relies on a flexible and reduced form estimation, rather than structural identification.

4.2 Clustering

We begin the empirical analysis by identifying broad heterogeneity in customer behavior using a clustering approach. Motivated by the economic framework, we cluster customers in a two-dimensional space defined by (i) prior belief, proxied by the smoothed historical open rate, and (ii) prior uncertainty, proxied by the variance of that open rate over a rolling window.

We apply K-means clustering, which is well suited to continuous, low-dimensional representations and yields economically interpretable partitions aligned with our theoretical constructs. Clustering variables are computed using only pre-sample customer history to avoid information leakage. Model tuning indicates a clear and stable three-cluster structure, supported by high silhouette scores³.

The resulting clusters can be naturally interpreted as *Loyal*, *Occasional*, and *Dormant* customers. As expected in mass marketing data, the majority of customers fall into the Dormant segment. Table 1 summarizes the size and defining characteristics of each group.

4.3 Empirical Model

To model the relationship between cognitive load, message characteristics, and opening decisions, we estimate a gradient-boosted decision tree model (XGBoost). This flexible framework allows for nonlinear effects and high-order interactions, which are central to our theory—particularly threshold effects in fatigue and heterogeneous responses across customer segments.

²For more detailed explanations of feature construction, underlying assumptions, and limitations, see Appendix A.

³Reported in Appendix B.

Table 1: Customer Segments Identified via Clustering

Cluster	Description	Number of Customers
Loyal	High prior belief, low uncertainty	39,000 (8%)
Occasional	Medium prior belief, high uncertainty	92,000 (19%)
Dormant	Low prior belief, low uncertainty	341,000 (72%)

Our objective is not to construct a purely predictive model, but to empirically map the theoretical mechanisms described in Section 3 to observed behavior. To emphasize generalization across individuals rather than temporal forecasting, we split the data at the customer level: the model is trained on one subset of customers and evaluated on unseen customers. This design ensures that model performance reflects structural patterns in behavior rather than customer-specific memorization.

Because the model is intentionally flexible, we rely on model-agnostic interpretability tools in subsequent sections to recover economically meaningful relationships between fatigue, signal features, and engagement.

4.4 Data and Modeling Assumptions

The following assumptions underpin our empirical approach:

1. **Proxy Validity:** Engineered features (e.g., historical open rates, subject line length, bonus keywords) are assumed to correlate with the underlying latent variables they intend to measure.
2. **Stationarity of Behavioral Patterns:** Customer responses to messaging features are assumed approximately stable within the observation window, allowing ALE plots and counterfactual simulations to provide meaningful insights.
3. **Partial Observability:** We assume that measurement errors in proxies do not completely obscure the latent patterns.
4. **Cluster Structure:** K-Means clustering on prior belief and uncertainty identifies meaningful segments; segment definitions are assumed stable over the observation period.

5 Results: Heterogeneity in Customer Responses

We examine heterogeneity in fatigue and attention responses across customer segments using cluster-specific Accumulated Local Effect (ALE) curves. ALE plots summarize the average local association between a feature and the model-predicted opening probability and are well suited to settings with correlated regressors and nonlinear models. This section evaluates whether the response patterns implied by the model align with the testable predictions derived in the theoretical framework.

5.1 Responses to Messaging Pressure

Figure 1 reports ALE curves for messaging pressure variables. We observe strong nonlinearities across all customer segments. Predicted opening probabilities increase with message volume at low

levels of exposure, but beyond a threshold of approximately two messages per week from the focal firm, the marginal association turns negative.

The decline is most pronounced among Occasional customers, reflecting a higher marginal sensitivity to fatigue-related proxies within this segment. Loyal customers exhibit a more moderate reduction, while Dormant customers display a gradual decline, indicating weaker marginal responses to additional exposure. These patterns are consistent with the model’s prediction that cognitive load generates non-linear disutility, with intermediate-belief customers exhibiting the strongest marginal sensitivity to incremental messaging pressure. Importantly, these differences reflect segment-specific responsiveness rather than mechanical differences in unconditional engagement levels.

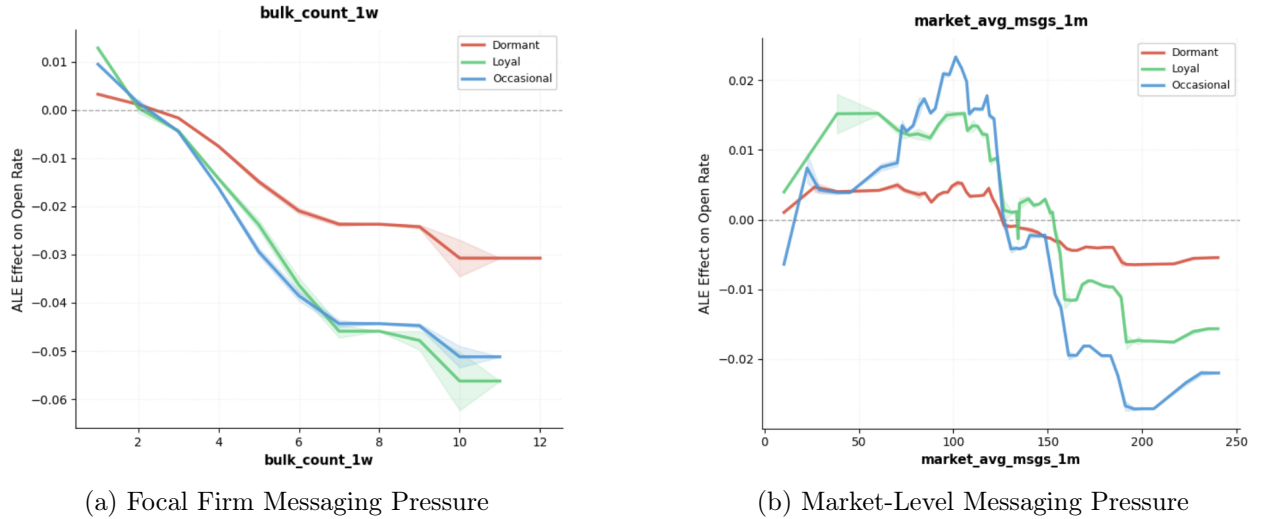


Figure 1: ALE Curves for Messaging Pressure by Customer Segment

5.2 Responses to Contextual and Content Features

Figure 2 presents ALE curves for contextual and content-related features. Temporal indicators such as weekend and afternoon delivery are positively associated with predicted engagement across all segments, with the strongest marginal responses observed among Occasional customers. Bonus-related subject line features similarly display a pronounced positive association for this group.

In contrast, Dormant and Loyal customers exhibit flatter response profiles, suggesting lower elasticity to short-term contextual and content variations. These patterns align with the theoretical prediction that contextual cues and message-level signals primarily influence customers with less precise or weakly held beliefs, while customers with more stable beliefs respond less to such transient features.

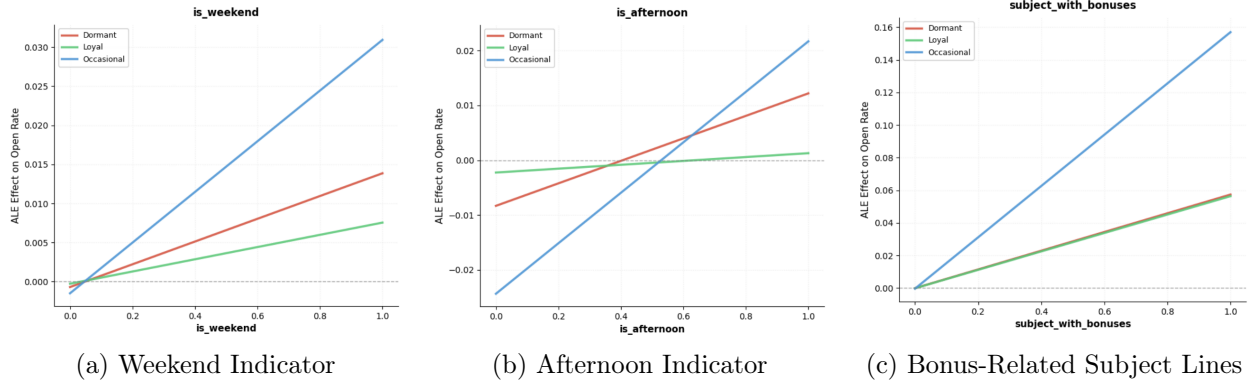


Figure 2: ALE Curves for Contextual and Content Features by Customer Segment

5.3 Summary of Segment-Level Responses

Taken together, the ALE results reveal substantial heterogeneity in how customer segments respond to messaging pressure and contextual features. Occasional customers exhibit the highest marginal sensitivity to both fatigue-related proxies and short-term contextual signals, consistent with their intermediate belief strength and lower prior precision. Dormant customers display comparatively muted responses across dimensions, while Loyal customers are more sensitive to messaging pressure but less responsive to signal content features. These differences reflect segment-specific marginal associations implied by the model and provide empirical support for the directional predictions of the theoretical framework.

These patterns reflect segment-specific marginal associations implied by the model and should not be interpreted as causal effects of messaging intensity or content.

6 Counterfactual Analyses

The counterfactual exercises presented in this section should be interpreted as model-based simulations of associations implied by the estimated predictive structure, rather than as policy-invariant causal interventions. Their purpose is to illustrate how the patterns identified in the data align with the mechanisms highlighted in our theoretical framework.

We consider two counterfactual settings. The first examines how message design interacts with fatigue, focusing on signal structure under different attention constraints. The second explores market-level coordination in messaging volume and its implications for engagement.

6.1 Signal Design Under Fatigue

Our first counterfactual evaluates how signal design performs under varying levels of fatigue. For each customer segment, we simulate predicted opening probabilities under combinations of: (i) low versus high messaging pressure, (ii) short versus long subject lines, and (iii) the presence versus absence of bonus-related keywords.

Consistent with our rational inattention framework, we expect richer signals to be effective when cognitive load is low, while simpler, high-salience signals perform better when attention is scarce. Longer subject lines contain more information but require greater processing effort, whereas shorter

subjects and bonus keywords increase the effective signal value while reducing evaluation costs.

Figure 3 summarizes the results. Under low fatigue, longer subject lines perform relatively well, particularly for Loyal customers. As fatigue increases, however, shorter subject lines and bonus-related keywords are associated with higher predicted opening probabilities across all segments, with the strongest response among Occasional customers. These patterns suggest that optimal signal design is state-dependent: message simplicity and salience become increasingly valuable as attention constraints tighten.

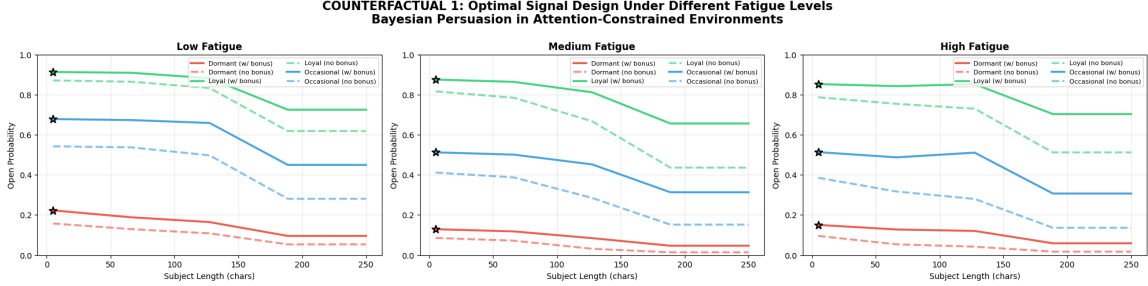


Figure 3: Signal design counterfactual under varying fatigue

6.2 Market Coordination and Messaging Volume

Our second counterfactual examines aggregate messaging pressure. When firms optimize message frequency independently, each ignores the congestion imposed on customers' attention, potentially leading to over-messaging. We simulate alternative coordination regimes by scaling aggregate messaging volume while holding other features fixed.

We compare five scenarios: (i) the status quo, (ii) unilateral self-restraint by a single firm (60% reduction), and (iii) industry-wide coordination with aggregate caps at 80%, 60%, and 40% of observed volume.

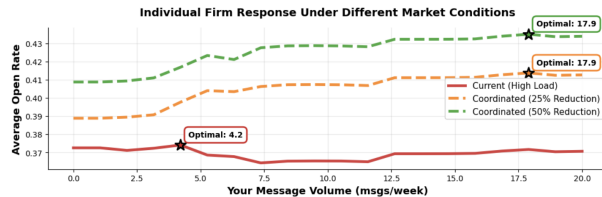
Table 2 reports predicted opening probabilities by segment. Unilateral self-restraint yields negligible improvements relative to the status quo. In contrast, coordinated reductions substantially increase predicted engagement across all segments, with the largest gains for Dormant and Occasional customers. Among coordinated policies, moderate reductions dominate extreme caps, indicating diminishing returns at very low message volumes.

Policy	Dormant	Loyal	Occasional	Average
Status Quo	0.061	0.733	0.277	0.357
Self-Restraint Only (60%)	0.061	0.739	0.278	0.359
Market Cap (80%)	0.073	0.763	0.301	0.379
Market Cap (60%)	0.082	0.790	0.339	0.404
Aggressive Cap (40%)	0.073	0.807	0.334	0.405

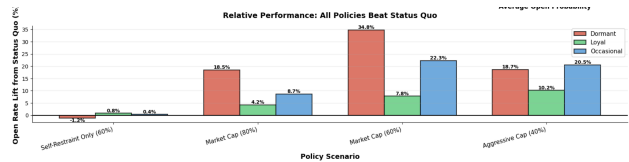
Table 2: Predicted Open Rates Under Alternative Coordination Regimes

Figure 4 visualizes these results. Lower aggregate messaging shifts predicted engagement upward, allowing firms to send messages more effectively in less crowded attention environments. These

patterns align with an “attention commons” interpretation: when firms coordinate to reduce total pressure, engagement improves for all, particularly for users most sensitive to fatigue.



(a) Predicted Engagement Under Different Aggregate Loads



(b) Engagement Lift Relative to Status Quo

Figure 4: Market Coordination Counterfactuals

7 Conclusion

In this paper, we explored how customer attention and fatigue influence engagement with marketing messages, using a framework grounded in bounded rationality and rational inattention. We operationalized theoretical concepts through machine learning models and cluster-specific analyses, demonstrating heterogeneous responses across customer segments.

Our analyses provide several key insights:

1. Customers exhibit distinct sensitivities to messaging volume and content: Occasional users are most responsive to short-term changes in signal and context, while Loyal and Dormant users display more stable behavior.
2. Signal design interacts with cognitive load: short, bonus-highlighted subject lines increase predicted engagement under higher fatigue, consistent with our theoretical predictions.
3. Market-level coordination in messaging can reduce collective fatigue and improve engagement, with the largest gains observed among Dormant and Occasional customers.

These results should be interpreted as associational rather than causal. All counterfactual exercises and model-based simulations reflect patterns implied by the estimated predictive structure. While informative for hypothesis generation and managerial intuition, they do not guarantee causal effects.

Limitations are acknowledged in multiple parts of the paper and summarized briefly here; for a detailed discussion, please see Appendix C. Key points include the reduced-form nature of the model, measurement error from proxies, anonymized data restricting feature richness, and sensitivity of cluster interpretations, especially for the Occasional segment.

Future Directions:

1. Develop dynamic models of belief updating to capture how opening behavior and content consumption influence posterior expectations.
2. Apply enhanced ML and econometric methods for more precise estimation of Bayesian updating under cognitive load.

3. Extend the framework to other datasets and contexts to assess external validity.
4. Incorporate macroeconomic or event data to account for short-term shocks affecting attention.
5. Investigate market coordination strategies using real measures of market pressure to simulate attention-constrained equilibria.

Overall, our study illustrates how theory-driven data analysis can guide marketing strategies under bounded rationality, while highlighting the exploratory nature of reduced-form and associational results.

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A Appendix A: Detailed Feature Construction

This appendix provides detailed descriptions of the engineered features used to approximate latent constructs from the economic framework.

A.1 Prior Belief and Prior Precision

- **Prior belief** (μ_i): calculated as the overall open rate of a user over their full previous history, smoothed using Bayesian shrinkage. This reflects persistent beliefs in the value of messages.
- **Prior precision** ($\tau_{\mu,i}$): computed as the variance of the user’s historical open rates over a rolling 14-day window. Low variance indicates high certainty in beliefs; high variance indicates greater uncertainty.

A.2 Signal Features

- **Subject line length**: shorter subject lines imply faster processing and higher effective precision.
- **Bonus keywords**: presence of promotional or economic value terms (e.g., bonuses) that increase perceived signal value.

A.3 Load and Cognitive Cost Proxies

- **Recent message volume**: total number of messages received by the user over 7-day and 30-day windows, capturing short-term and more persistent fatigue effects.
- **Simulated competing messages**: actual competitor-specific data is unavailable; we simulate market-level messaging pressure using overall company message volumes as a proxy.
- **Time-of-day and day-of-week indicators**: capture periodic variation in attention and global load pressures.
- **Event-type proxies**: global open rates over the previous week and month capture country- or platform-level fluctuations (e.g., macroeconomic shocks, COVID pandemic, war periods) that may affect overall engagement.

A.4 Measurement Error and Identification

All engineered features are proxies for latent variables and include measurement error. For example:

- Prior belief may not perfectly capture persistent expectations due to smoothing and limited history.
- Prior precision is based on observed variance in open rates over 14-day windows, which may be noisy for users with few messages.
- Signal features may not perfectly capture perceived informativeness due to subjective user preferences.
- Load proxies capture only observable components; unobserved fluctuations are modeled as ϵ_{it}^L .

The combination of idiosyncratic shocks (ϵ_{it}^{EV} , ϵ_{it}^L) and measurement error motivates our flexible reduced-form estimation, rather than relying on structural identification.

B Appendix B: Clustering and Model Diagnostics

B.1 Clustering Diagnostics

Figure 5 visualizes the customer clusters in the belief–uncertainty space. The separation between clusters aligns closely with the economic interpretation of persistent beliefs and attention allocation.

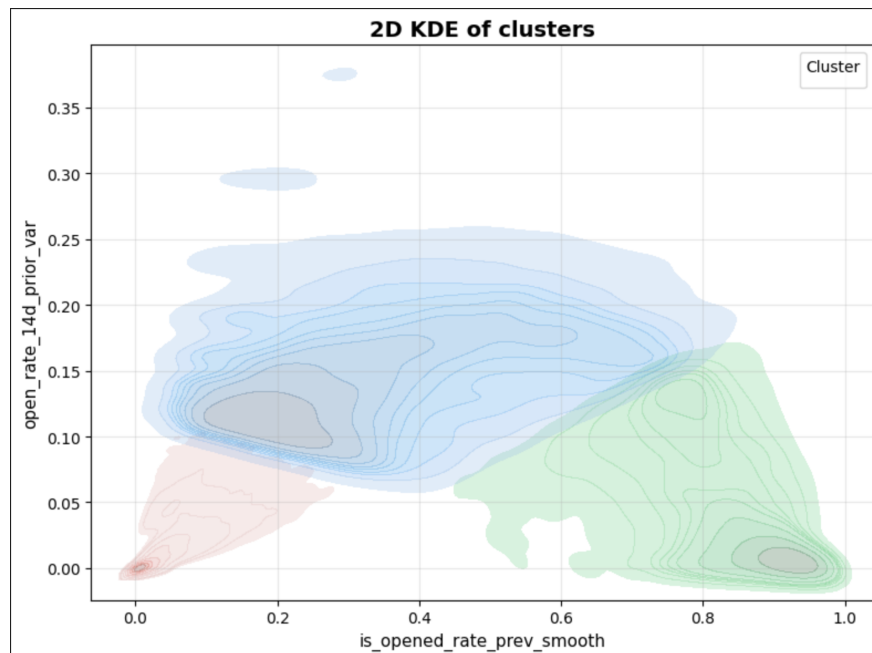


Figure 5: Customer clusters in the belief–uncertainty space.

Figure 6 reports silhouette scores across alternative numbers of clusters. A three-cluster solution maximizes average silhouette width, indicating a well-defined partition.

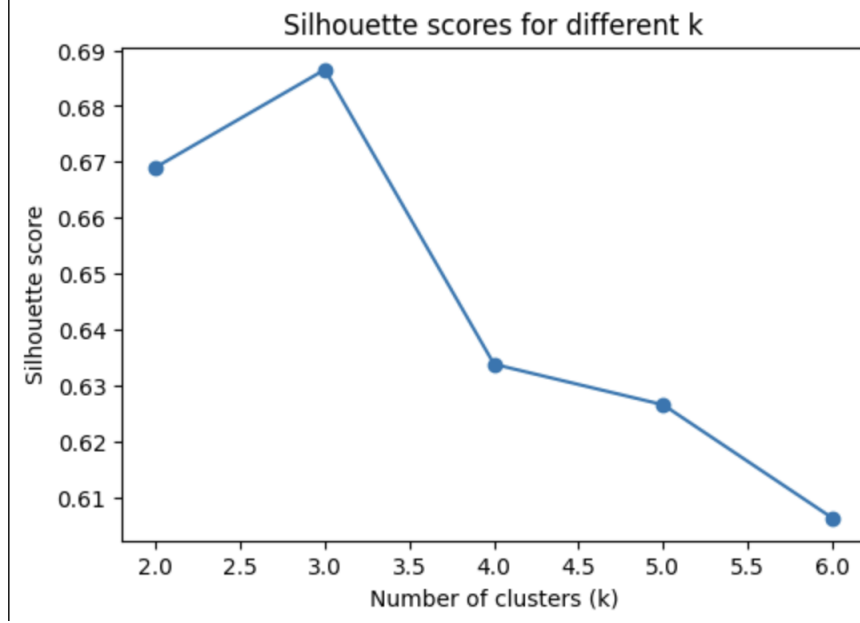
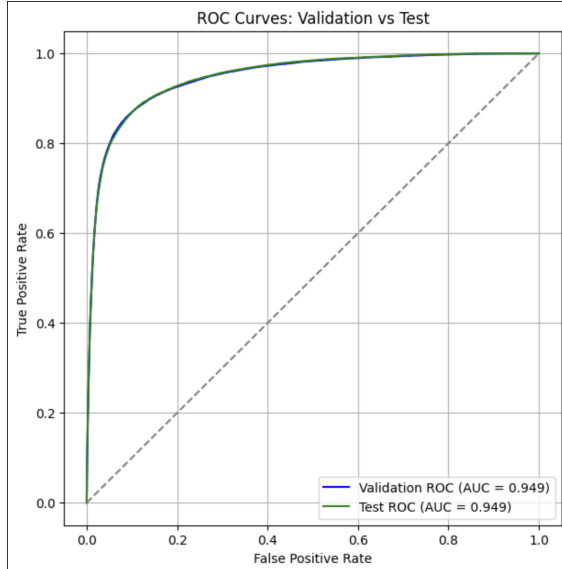


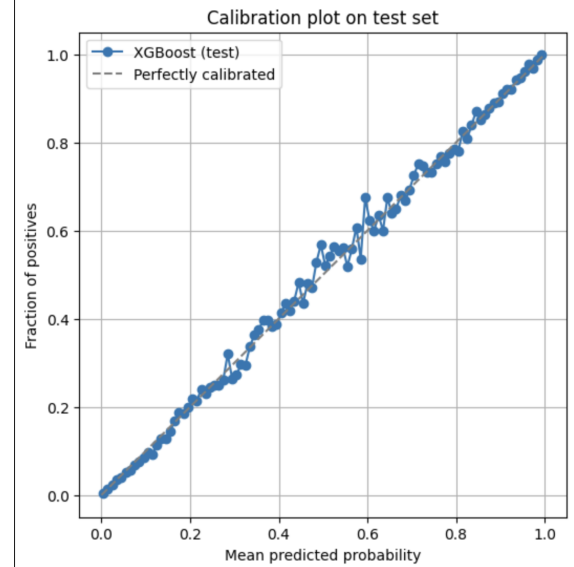
Figure 6: Silhouette scores across different numbers of clusters.

B.2 Predictive Performance and Calibration

Although prediction is not the primary objective, we report standard performance diagnostics to assess model reliability. Figure 7a shows the ROC curve for the held-out customer sample, and Figure 7b presents calibration plots comparing predicted and observed opening probabilities.



(a) ROC curve for the held-out customer sample.



(b) Calibration plots comparing predicted and observed opening probabilities.

C Detailed Limitations

The following limitations apply to our analyses, both for ALE plots and counterfactual simulations:

1. **Associational Nature of Results:** All model predictions and counterfactual exercises reflect correlations, not causal effects. Observed associations may be influenced by unobserved targeting, seasonality, competition, or customer heterogeneity.
2. **Reduced-Form Modeling:** XGBoost captures complex interactions, but it does not enforce structural economic parameters (costs, capacities, precision). Interpretations through the rational inattention framework are therefore indirect.
3. **Measurement Error from Proxies:** Features such as prior belief, prior precision, load, and signal strength are proxies and imperfect. Measurement error may attenuate estimated associations or introduce bias.
4. **Anonymized Data Limitations:** The dataset lacks personal or product-level information, limiting our ability to measure customer-specific preferences, seasonality, or loyalty dynamics. More detailed firm-level data would enhance precision and structural inference.
5. **Cluster Sensitivity:** The Occasional segment may include seasonal or context-dependent customers. High sensitivity to fatigue proxies may partly reflect underlying seasonality or intermittent interest rather than true attention fatigue.
6. **Sparse or Extreme Feature Spaces:** Some counterfactual scenarios explore rarely observed combinations of features (e.g., extreme fatigue with atypical subject line design). Predictions in these regions rely on extrapolation and should be interpreted cautiously.
7. **Limited Observation Period:** Only two years of data are available, covering unusual events such as the COVID-19 pandemic and geopolitical shocks. Short observation windows may limit the ability to identify recurring patterns or fully disentangle short-term effects from seasonal behavior.

These limitations emphasize that our results are illustrative and exploratory, offering guidance for managerial intuition rather than definitive policy prescriptions.